

TP04: Comparing KNN and SVM

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1 Introduction

K-Nearest Neighbors (KNN) is a non-parametric algorithm that classifies data points based on the majority class of their k nearest neighbors in the feature space. It makes predictions by measuring distances between points.

Support Vector Machine (SVM) is a supervised learning algorithm that finds an optimal hyperplane to separate classes by maximizing the margin between them. Different kernel functions allow SVM to handle non-linear decision boundaries.

2 Code Execution & Verification

The algorithms have been successfully executed on the `Social_Network_Ads` dataset. The dataset contains 400 samples with 2 features (Age, EstimatedSalary) predicting whether a customer purchased a product (binary classification).

Dataset Details:

- Total samples: 400
- Features: Age, EstimatedSalary
- Target classes: 0 (not purchased) = 257, 1 (purchased) = 143
- Training set: 300 samples — Test set: 100 samples

3 Data Preprocessing Analysis

3.1 Applied Preprocessing: Feature Scaling (Standardization)

Before scaling: Age range: 18–60, Salary range: 15,000–150,000

After scaling: Both features have mean = 0.0, std = 1.0, range [-2.5, 2.5]

3.2 Why Feature Scaling is Necessary

Feature scaling is critical because Age and Salary have vastly different scales. Without scaling, the Salary feature (ranging in thousands) would dominate distance calculations in KNN and affect SVM's optimization process, leading to biased predictions toward the higher-magnitude feature and poor decision boundaries.

4 Performance Evaluation

4.1 Confusion Matrix Analysis

Algorithm	Confusion Matrix	Accuracy
KNN (k=5)	$\begin{bmatrix} 64 & 4 \\ 3 & 29 \end{bmatrix}$	93%
SVM (Linear)	$\begin{bmatrix} 66 & 2 \\ 8 & 24 \end{bmatrix}$	90%
SVM (Polynomial)	$\begin{bmatrix} 67 & 1 \\ 13 & 19 \end{bmatrix}$	86%
SVM (RBF)	$\begin{bmatrix} 64 & 4 \\ 3 & 29 \end{bmatrix}$	93%

Table 1: Confusion matrices showing TN, FP, FN, TP (row format)

4.2 Algorithm Comparison

Best Performers: KNN and SVM with RBF kernel tied at 93% accuracy with identical confusion matrices.

Why KNN Performs Well: The dataset has non-linear, localized patterns where similar customers cluster together. KNN naturally captures these local patterns without assuming a specific decision boundary shape.

Kernel Comparison: RBF kernel matched KNN's performance by capturing non-linear patterns. Linear kernel (90%) suggests data isn't perfectly linearly separable. Polynomial kernel (86%) struggled with 13 false negatives, indicating overfitting.

5 Decision Boundary Visualization

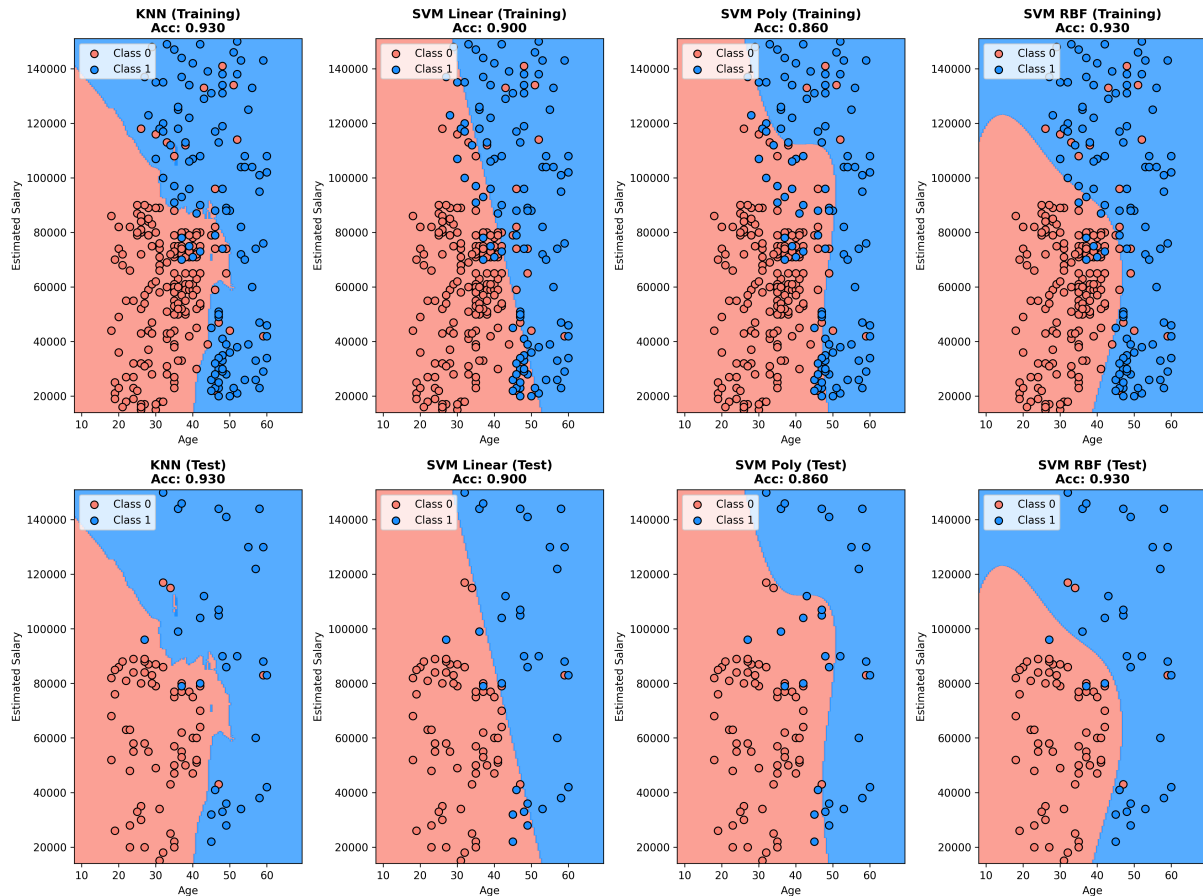


Figure 1: Decision boundary comparison for KNN and SVM variants on training and test sets

5.1 Interpretation of Decision Boundaries

KNN (k=5): Creates smooth, irregular boundaries that adapt to local data clusters, following natural grouping patterns.

SVM Linear: Produces straight-line separation. While achieving 90% accuracy, the linear boundary cannot capture the curved pattern, leading to misclassifications.

SVM Polynomial: Creates complex, curved boundaries but overfits in sparse regions, resulting in the lowest accuracy (86%).

SVM RBF: Produces smooth, curved boundaries similar to KNN, effectively capturing the non-linear pattern and matching KNN's 93% accuracy.

6 Bonus: Hyperparameter Tuning

6.1 Experimental Results

KNN - k values:

- k=1: 87% (overfitting to noise)
- k=3 to k=20: 93% (stable performance)

SVM RBF - C values:

- $C=0.1$: 92%
- $C=1$ to $C=100$: 93% (optimal)

SVM RBF - Gamma values:

- $\text{gamma}=0.01$: 89% (underfitting)
- $\text{gamma}=0.1$: 92%
- $\text{gamma}=1$ to $\text{gamma}=10$: 93% (optimal)

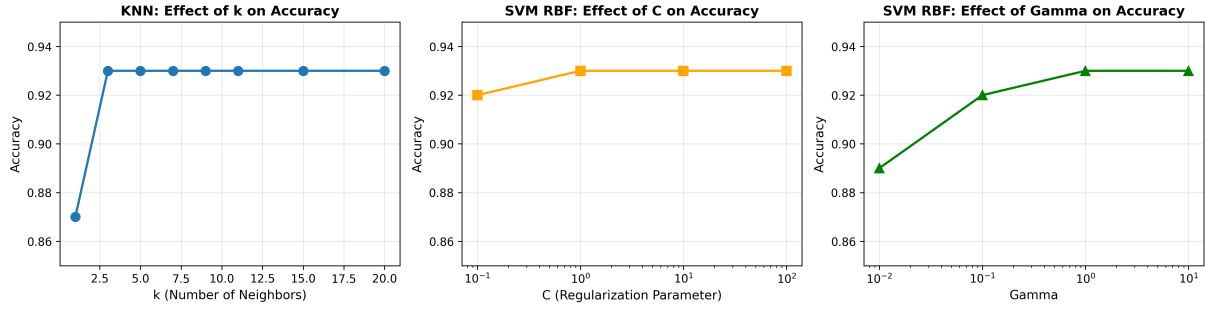


Figure 2: Effect of hyperparameter tuning on model accuracy

6.2 Sensitivity Analysis

KNN: Highly sensitive to k when very small ($k=1$ shows 6% accuracy drop). However, stable across $k=3$ to $k=20$, indicating robustness once k exceeds a minimum threshold.

SVM: More sensitive to gamma than C . Low gamma (0.01) causes underfitting with overly smooth boundaries. Both parameters show stable performance once optimal values are reached.

7 Conclusion

KNN ($k=5$) and SVM with RBF kernel achieved the highest accuracy of 93% with identical confusion matrices. KNN excels due to the dataset's non-linear, localized patterns where similar customers cluster together. The RBF kernel matched this by mapping data to higher dimensions, while Linear SVM (90%) struggled with non-linearity and Polynomial SVM (86%) overfitted. Feature scaling was essential given the vast scale difference between Age and Salary. Hyperparameter tuning revealed KNN's stability for $k=3$ and SVM's sensitivity to gamma values. Both algorithms are viable for this dataset, with KNN offering simplicity and SVM-RBF providing theoretical guarantees.